Name: ___________________________

Starter quiz

Multiples of 10 involving negative exponents

1 A prime number is an integer greater than one with exactly ______________________ factors. Fill in the blank

2 Which of the following products only use prime numbers? Tick **2** correct answers

- ☐ $2 \times 2 \times 13$
- ☐ $2 \times 6 \times 13$
- ☐ $2 \times 9 \times 13$
- ☐ $2 \times 3 \times 5 \times 17$

3 Write 300 as a product of its prime factors. Tick **1** correct answer

- ☐ $4 \times 3 \times 25$
- ☐ $2^2 \times 3 \times 5^2$
- ☐ $4 \times 3 \times 5^2$
- ☐ $2^2 \times 3 \times 25$
- ☐ $2^2 \times 3 \times 5$

4 Use the fact that $60 = 2^2 \times 3 \times 5$ to write 300 as a product of its prime factors. Tick **1** correct answer

- ☐ $2^2 \times 3^2 \times 5$
- ☐ $2^2 \times 3 \times 5^2$
- ☐ $2^2 \times 3^2 \times 5^2$
- ☐ $2 \times 3 \times 5^2$

5 Use the fact that $60 = 2^2 \times 3 \times 5$ to write 30 as a product of its prime factors. Tick **1** correct answer

- [ ] $2^2 \times 3 \times 5 \div 2$
- [ ] $2 \times 3 \times 5$
- [ ] $2^2 \times 5$
- [ ] $2 \times 3 \times 5^2$

6 If $52 = 2^2 \times 13$ and $175 = 5^2 \times 7$, which of the following is the most useful arrangement to calculate 175×52? Tick **1** correct answer

- [ ] $2 \times 2 \times 13 \times 5 \times 5 \times 7$
- [ ] $2 \times 13 \times 2 \times 5 \times 5 \times 7$
- [ ] $2 \times 5 \times 2 \times 5 \times 13 \times 7$
- [ ] $2 \times 7 \times 2 \times 5 \times 13 \times 5$